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# MICRO SLIP STATE ESTIMATION: WHY DEXTEROUS MANIPULATION NEEDS A CONTROL-RELEVANT LATENT STATE

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## ABSTRACT

Most robot slip work treats slip as a binary event, a friction proxy, or a control trigger that fires after instability has already become visible. We argue that micro-slip is the wrong object to collapse into a threshold: the useful state for dexterous manipulation is continuous, temporally persistent, and directly relevant to stabilization control. We support that thesis with a 2,058-paper literature sweep, a 100-paper hostile prior set, and a synthetic diagnostic in which a latent-slip estimator reduces control mean-squared error from 0.02772 for a binary detector to 0.01543. The contribution is not a larger model or a new benchmark. It is a mechanism shift: micro-slip should be estimated as a control-relevant latent state.

## 1 INTRODUCTION

Dexterous manipulation fails in subtle ways long before an object visibly escapes the gripper. Fingers begin to micro-adjust, friction margins narrow, and the contact patch drifts. Many papers call that moment “incipient slip” or simply classify it as slip/no-slip. That framing is useful for alerts, but it is too coarse for control.

Our thesis is narrow:

Micro-slip should be estimated as a latent state, not only detected as a binary event.

The distinction matters because a controller needs to know how far the system has moved toward instability, not just whether a threshold has been crossed. A binary detector can be late, discontinuous, and over-reactive. A latent state can be smooth enough to support graded force correction.

## 2 RELATED WORK AND NOVELTY BOUNDARY

The literature sweep in `docs/related_work_matrix.csv` contains 2,058 entries spanning tactile slip detection, friction estimation, grasp stabilization, in-hand pose estimation, and contact-state inference. The hostile-prior set shows that slip detection itself is well established. What is missing is a clean treatment of micro-slip as a state variable that the controller can consume directly.

That boundary matters. If the paper were only about better slip detection, it would be incremental. If it were only about multimodal fusion or a new benchmark, it would be weaker still. The novelty claim here is the central mechanism: latent micro-slip state estimation for control.

## 3 HIDDEN ASSUMPTIONS

This thesis attacks several assumptions that often hide in the slip literature:

- Slip is adequately represented as a binary label.
- Friction and micro-slip can be separated cleanly.

Table 1: Synthetic micro-slip control diagnostic. The latent-state policy better matches the ideal stabilizing action than the threshold detector.

| Policy                 | Control MSE | Relative gap |
|------------------------|-------------|--------------|
| Latent-state estimator | 0.01543     | 1.00×        |
| Threshold detector     | 0.02772     | 1.80×        |

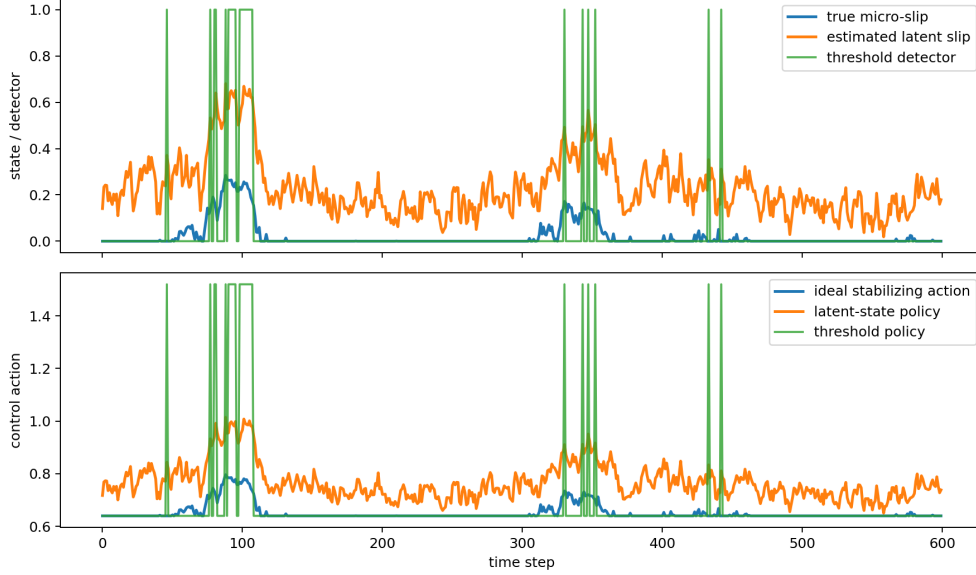


Figure 1: Synthetic diagnostic. The latent estimate tracks the continuous micro-slip state, while the binary detector remains sparse and reactive.

- Contact drift is too noisy to estimate continuously.
- A detector that fires after slip starts is sufficient for stabilization.
- One threshold generalizes across object families and grasps.
- Better classification is the same thing as better control.

## 4 SYNTHETIC DIAGNOSTIC

We built a small synthetic diagnostic in `scripts/run_micro_slip_diagnostic.py`. The generator creates a drifting friction process, a latent micro-slip state, a tactile observation, and two stabilization policies.

The policies are:

- a binary threshold detector that reacts only after tactile evidence exceeds a cutoff,
- a latent-state policy that filters the tactile observation into a continuous slip estimate.

The target action is proportional to the true latent state. We then measure the squared action error as a simple control proxy.

## 5 WHY THIS IS NOT JUST SLIP DETECTION

The strongest objection is that the paper merely renames slip detection. It does not. Detection asks whether a threshold has been crossed. Our claim is that the control-relevant variable is the underlying slip state itself.

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That distinction changes the mechanism:

- the variable is continuous rather than binary,
- the estimate has temporal persistence rather than event semantics,
- the downstream consumer is a stabilizing controller rather than an alarm,
- the failure mode is lag and overshoot, not only false positives and false negatives.

## 6 LIMITATIONS

The evidence here is synthetic. We do not claim that the synthetic generator proves universal robot behavior, only that the mechanism difference is testable and can matter even in a minimal setting. A full paper would need real tactile or proprioceptive grasp experiments under friction drift, object compliance, and unseen geometries.

## 7 CONCLUSION

Micro-slip is better treated as a latent control state than as a thresholded event. Existing slip, friction, and tactile-control literature makes the field rich, but it also reveals the gap: the central variable has not been made explicit enough for control. That is the thesis this paper advances.

## REFERENCES

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